# How Effective are Coverage- and Diversity-Based Test Selection at Killing Stubborn Mutants?

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***Abstract*—Coverage- and diversity-based test selection are two prominent strategies for accelerating regression testing by deriving a smaller set of tests from an original test suite. While coverage-based approaches maximise fault detection, they require up-front code instrumentation and execution. Diversity-based approaches, on the other hand, select tests based on dissimilarity, but it has been found that they discover fewer faults. However, the specific characteristics of the faults detected by these two strategies remain unexplored. This raises critical questions: Does the superior raw performance of coverage-based approaches stem from merely finding "trivial" faults, and how do both strategies fare against subtle, hard-to-detect bugs? This is vital because optimising for raw fault counts alone may create a false sense of security, masking an inability to detect deeper faults that could present unique challenges to system reliability. Mutation analysis offers a means of studying this behaviour, simulating hard-to-find bugs by generating "stubborn" mutants — mutants that can only be detected by a small number of tests. We empirically investigated how coverage- and diversity-based approaches perform with mutants and real faults of varying stubbornness across seven Java projects. Results show that diversity-based selection often rivals or surpasses coverage for the most difficult mutants, requiring up to 46% fewer tests. We further show that tests killing only stubborn mutants constitute just 2–25% of a suite yet achieve full mutation adequacy and detect up to 88% of real faults, and that such mutants are disproportionately found in private, void, and deeply nested methods. These findings highlight stubborn mutants as efficient test targets and lightweight diversity-based selection as a scalable alternative to coverage, avoiding pre-execution and instrumentation costs.**

***Index Terms*—mutation testing, stubborn mutant, diversity-based testing, test case selection**

## I. Introduction

Regression testing ensures that software modifications do not introduce new faults. As projects evolve, test suites grow in size and execution cost, often making it impractical to run every test after every change [32]. To address this, researchers have developed test selection strategies to execute only the tests deemed most valuable — for example, by finding the most faults, or faults that other tests are unlikely to detect. *Coverage-based* techniques address this problem by selecting tests that maximise coverage. While empirical studies show that they consistently find the most faults [29], [32], they require instrumentation of project code and pre-execution of the entire test suite to collect coverage information, which can be costly for large systems. *Diversity-based* approaches, in contrast, select tests that are most dissimilar to others in the test suite, using a similarity metric [12]. Typically, diversity-based approaches detect fewer faults, but are more lightweight, since diversity can be measured statically using the source code [21], [9] or bytecode [10] of the tests.

Despite these apparent trade-offs, the *nature* of the faults detected by these approaches remains unexplored. For instance, while coverage-based techniques yield higher fault counts, they may primarily uncover trivial defects exposed by simple code execution. This is problematic: favouring raw detection counts could create a false sense of security for developers, and emphasising these tests over those capable of exposing deeper, subtle faults risks compromising the reliability of future system releases.

This paper is the first to investigate this issue, applying mutation analysis [17] to generate artificial faults of varying "difficulty". Particularly hard-to-detect faults are referred to as *stubborn* mutants. Figure 1 shows an example from the `Fraction` class in the widely-used Apache Commons Math library for Java, and one of our subjects used in our experimental analysis later in the paper, with three mutants depicted. The first mutant ($M_1$) affects line 5 and is detected ("killed") by almost all tests, because it reverses a rare overflow condition that throws an exception. The second ($M_2$) affects line 8. It affects the calculation of a value used in the condition, and is consequently more "difficult" to detect and is killed by fewer tests. Finally, the third ($M_3$) alters line 38, and is only killed by a very small number of tests. This mutant introduces an "off-by-one" error, and is particularly hard to detect. To do so, requires a very specific input combination that makes two values equal to one another, thereby exposing the difference between the mutant and the original version of the code.

The relationship between *test selection* and *mutant difficulty* has not been systematically examined. Existing studies mainly report overall mutation scores or fault detection rates, implicitly treating all mutants as equally important [21], [24], [10]. Consequently, it remains unclear whether coverage-based techniques perform well because they detect *many* faults or because they detect the *most difficult* ones. Clarifying this distinction is essential for assessing the true value of different selection strategies, especially in cost-sensitive settings such as regression testing and continuous integration. Because stubborn mutants correlate with real fault difficulty [19], [27], they also offer a conceptual bridge to understanding "hard-to-find" real faults (i.e., faults that are killed by very few tests). In this work, we extend this connection by analysing not only stubborn mutants but also real faults that are difficult to detect.

```
1   private Fraction(double value, double epsilon,
        int maxDenominator, int maxIterations) {
2     long overflow = Integer.MAX_VALUE;
3-4   double r0 = value; long a0 = (long)floor(r0);
5   ⊖ if (a0 > overflow) {
    ⊕ if (a0 <= overflow) { // M1
6       throw new FractionException(value, a0, 1l);
7     }
8   ⊖ if (abs(a0 - value) < epsilon) {
    ⊕ if (abs(a0 + value) < epsilon) { // M2
9       this.numerator= (int) a0;
10-11   this.denominator = 1; return;
12    }
      ...
23-26 long p2=...;long q2=...;int n=...;
27    do {
        ...
37      double convergent = (double)p2/(double)q2;
38  ⊖   if(n<maxIterations && abs(convergent-value)
            > epsilon && q2<maxDenominator) {
    ⊕   if(n<maxIterations && abs(convergent-value)
            > epsilon && q2<=maxDenominator) { // M3
        // ...
47    } while (...);
      ...
    }
```

Figure 1: A constructor of the `Fraction` class from the Apache Commons Math library, demonstrating three mutants, M1–M3. Omitted code sections are replaced with "`...`".

To address this gap, we analyse how test selection strategies perform across mutants of varying stubbornness. We compare coverage-based and diversity-based approaches on seven real-world Java projects from Defects4J [18], examining: (1) their effectiveness in killing stubborn mutants; (2) the structural and semantic characteristics of these mutants; and (3) the practical value of the tests that kill them. Results show that diversity-based using bytecode information (i.e., the compiled form of tests) performs comparably or better than coverage-based selection, requiring 46% fewer tests on median to kill the most stubborn mutants. Also, diversity-based required 11–40% fewer tests than coverage-based selection to detect the hardest faults. Stubborn mutants are disproportionately found in private and `void` methods, deeply nested code, and are frequently produced by *method-level* mutation operators (e.g., `VoidMethodCall`). These findings provide new insight into where and how stubborn mutants occur and can guide the design of more effective mutation operators targeting high-utility faults. Finally, we show that tests killing only stubborn mutants constitute just 2–25% of a suite, yet achieve full mutation adequacy and detect up to 88% of real faults.

These findings provide the first empirical evidence that diversity-based test selection can effectively detect hard-to-find faults, while also deepening our understanding of their properties and practical significance. We provide a replication package [1] containing all data, source code, and instructions for reproducing our experiments. This paper makes the following contributions:

1) The first empirical study of how test selection strategies relate to fault difficulty, using stubborn mutants and hard-to-find real faults.
2) A comparison of coverage-based and diversity-based test selection on seven real-world projects, analysing effectiveness across mutant stubbornness levels.
3) An empirical characterisation of stubborn mutants' code locations, mutation operator types, and their practical relevance for mutation testing.

## II. Background

### A. Mutation Analysis

Mutation analysis is a fault-based technique that evaluates test suite effectiveness by introducing small syntactic changes, called *mutants*, into a program [17], simulating real developer faults. It relies on two hypotheses: the Competent Programmer Hypothesis (CPH) [7], which assumes programmers write nearly correct code with small deviations, and the Coupling Effect [25], which suggests that tests detecting simple errors also detect more complex ones.

A mutant is *killed* if at least one test fails on the mutated code, and *live* if all tests pass, indicating potential weaknesses in the developer test suite. Some mutants can never be killed; these are *equivalent mutants*, which are syntactically different but semantically identical to the original program.

### B. Stubborn Mutants

In principle, the idea of a stubborn mutant is the same across different works, but the definition of these stubborn or hard-to-kill mutants varies. A stubborn mutant is one that is very hard to kill, while an equivalent mutant is impossible to kill. All mutants lie on a spectrum of "killability" from easy to kill to impossible to kill [31]. Stubborn mutants occupy the space immediately adjacent to equivalent mutants on this spectrum. They subsume easier-to-kill mutants [20], [23], making them a stronger indicator of test suite effectiveness, and their difficulty correlates more strongly with real fault detection [19], [27].

Papadakis et al. [27], Hernandez et al. [15], and Du et al. [8] consider a mutant to be stubborn if very few tests kill that mutant. Hernandez et al. [15] introduced the rank-stubbornness model (RSM), where they evaluate the stubbornness rank of each mutant. The RSM assigns each mutant a rank based on the number of test cases that kill it. Mutants killed by fewer test cases are assigned higher ranks, with those killed by only one test case receiving the highest rank. We will refer to the RSM later in Section III-A. Also, Lindström et al. [23] reported that a stubborn mutant is one that is killed only by one test or very few tests. Our work uses a similar definition of stubborn mutants.

### C. Test Selection Techniques

Test selection techniques aim to identify a subset of test cases from a larger suite that maximises fault detection while reducing execution cost [32]. These techniques are widely used in regression testing and continuous integration, where executing the full suite can be expensive.

Coverage-based techniques select tests based on the program elements (e.g., statements, branches, or methods) they

Table I: Example showing mutant stubbornness and kill scores

| **Tests** | $m_1$ | $m_2$ | $m_3$ | $m_4$ | $m_5$ | $m_6$ | $m_7$ |
|---|---|---|---|---|---|---|---|
| $T_1$ | ✓ | ✓ | ✗ | ✓ | ✗ | ✗ | ✓ |
| $T_2$ | ✓ | ✗ | ✗ | ✓ | ✗ | ✗ | ✗ |
| $T_3$ | ✓ | ✗ | ✗ | ✓ | ✓ | ✗ | ✓ |
| $T_4$ | ✓ | ✓ | ✓ | ✗ | ✗ | ✓ | ✗ |
| $T_5$ | ✓ | ✓ | ✗ | ✗ | ✗ | ✗ | ✗ |
| $T_6$ | ✓ | ✗ | ✗ | ✓ | ✓ | ✓ | ✓ |
| $T_7$ | ✗ | ✓ | ✗ | ✓ | ✗ | ✗ | ✓ |
| **Kill score** | 6 | 4 | 1 | 5 | 2 | 2 | 4 |

exercise [29]. These approaches are *dynamic*, requiring pre-execution of the test suite to collect coverage information, which can be costly for large systems. Despite this, coverage-based selection has consistently been shown to achieve strong fault detection performance.

Diversity-based techniques, in contrast, select tests based on *dissimilarity* between test cases [12]. Dissimilarity is measured using a similarity metric (e.g., Euclidean distance, or Jaccard similarity) over program artefacts, such as test inputs or outputs. These methods are *static* and *lightweight*, as they do not require test execution, making them substantially cheaper than coverage-based strategies [10].

## III. Approach

This section outlines our approach to identifying and evaluating stubborn mutants and analyses how different test selection strategies detect them early. We present our model for identifying stubborn mutants and describe the test selection strategies employed in our study. Then, we assess the significance of the tests killing only stubborn mutants.

### A. Stubborn Mutant Identification

Table I shows an example of seven tests and seven mutants. Each cell entry indicates whether a test $T_i$ kills a mutant $m_j$ ('✓') or not ('✗'). As discussed in Section II-B, the Rank-Stubbornness Model (RSM) assigns a rank to each mutant based on how many tests kill it. However, the term rank may be misleading, as it can suggest that each mutant is assigned a contiguous ordinal position. Instead, we use the term **Kill Score**, representing the number of distinct test cases that kill each mutant, allowing multiple mutants to share the same score. The last row of Table I (*Kill Score*) shows these values. For instance, mutant $m_1$ is killed by six tests, indicating it is easy to detect, while $m_3$ is killed by only one test and $m_5$ and $m_6$ by two, making them harder to kill.

The Rank-Stubbornness Model (RSM) ranks mutants by their resistance to being killed but lacks a principled threshold for deciding which mutants are truly stubborn, relying instead on arbitrary cutoffs. Papadakis et al. [27] introduced fixed thresholds (e.g., mutants killed by no more than 5% or 2.5% of tests are considered stubborn). However, such rigid cutoffs can misclassify mutants, particularly in projects where even easy mutants are killed by few tests. In such cases, many mutants may be incorrectly labelled as stubborn.

Stubbornness is inherently project-dependent: a score that denotes high resistance in one system may represent average difficulty in another. Thus, absolute thresholds yield inconsistent or misleading classifications across projects.

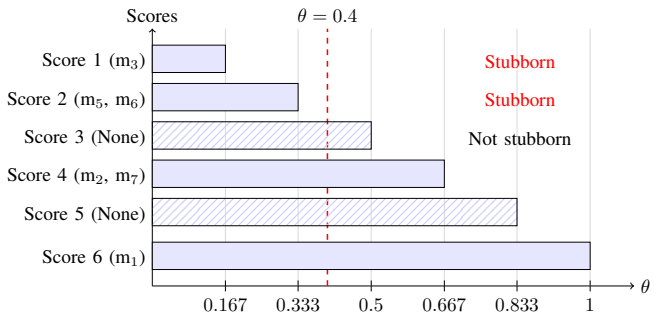



Figure 2: Illustration of RSTM: grouping mutants by threshold levels. Filled bands represent mutants present in the example; hatched bands (Score 3 and 5) are added for illustration but contain no mutants in this scenario.

To address these issues, we propose the **Relative Stubbornness Thresholding Model (RSTM)**. RSTM introduces a user-defined relative threshold that identifies stubborn mutants based on their difficulty *relative to the easiest mutant* within the same project. This approach enables consistent, flexible, and project-aware identification of stubborn mutants. By adjusting the threshold, testers can control the sensitivity of stubbornness detection to suit their analysis needs.

For a set of $n$ mutants and a test suite $T$, RSTM assigns each mutant $m_i$ a Kill score $s_i \in \mathbb{N}$, which is the number of test cases in $T$ that kill $m_i$. Let:

- $S = \{s_1, s_2, \ldots, s_n \mid 1 \leq s_i \leq |T|\}$ be the multiset of Kill scores assigned to each of the $n$ killable mutants.
- $s_{\max}$ be the highest score value, corresponding to the easiest-to-kill mutant (i.e., killed by the most tests).
- $\theta \in (0, 1]$ be a user-defined threshold that determines the level of relative stubbornness.

A mutant $m_i$ is classified as *stubborn* under the RSTM if:

$$s_i \leq \theta \cdot s_{\max}$$

Using the example from Table I, $s_{\max} = 6$ and if $\theta = 0.4$, then the threshold is $6 \times 0.4 = 2.4$, and any mutant with $s_i \leq 2.4$ is considered stubborn. So, $m_3$, $m_5$, and $m_6$ are stubborn mutants. If the stubbornness threshold is 0.25, then only $m_3$ is stubborn. RSTM enables flexible identification of relatively stubborn (i.e., harder-to-kill) mutants.

Figure 2 visualises this example. The y-axis shows the mutant scores (number of killing tests), while the x-axis represents the corresponding threshold levels ($\theta$). Mutants are grouped into horizontal bands by score, and the $\theta$ line defines which groups fall below the threshold. For instance, with $\theta = 0.4$, mutants in groups with $\theta \leq 0.4$ (e.g., $m_3$, $m_5$, $m_6$) are classified as stubborn, while those in higher bands are not.

To extend this analysis from artificial faults to real faults, we applied the same RSTM logic to each real fault in our dataset. Specifically, we treated each real fault analogously to a mutant by considering the number of test cases that detect it. A real fault is classified as *hard-to-find* if the number of tests

detecting it ($s_i$) falls below the relative stubbornness threshold (i.e., $s_i \leq \theta \cdot s_{\max}$), where $s_{\max}$ corresponds to the easiest-to-kill mutant within the same project. This approach enables a unified and project-aware characterisation of fault difficulty for both mutants and real faults, allowing us to directly compare how test selection strategies perform on hard-to-find faults in both categories.

### B. Test Case Selection Approaches

We implemented six test case selection approaches, two of which are coverage-based and four are diversity-based.

The first coverage-based selection approach is based on maximising statement coverage [29], as the hypothesis could be that stubborn mutants are killed by comprehensive, broadly-scoped tests that cover large portions of the code. We implemented a greedy algorithm where the test case with the maximum coverage is selected first. If multiple test cases have the same maximum coverage, the first one is selected. Then, the test case with the next highest *additional* coverage is selected in the same way, and this continues until the stubborn mutant under consideration is killed. Since all stubborn mutants in our study are by definition killable (as identified by RSTM), this process always terminates successfully. The second coverage-based selection approach is based on minimum statement coverage. The selection is exactly as before, except that the minimum coverage is used to guide the selection process. We did this to investigate whether tests with a narrow, focused coverage have any effect on killing stubborn mutants over tests that have a wider focus. By implementing these two opposing strategies, we can analyse which coverage-based property (breadth or focus) is more strongly associated with the ability to reveal stubborn mutants.

We implemented two state-of-the-art diversity-based selection approaches, Ledru [21] and FAST [24], motivated by the idea that stubborn mutants are hard to kill and may require diverse test inputs or behaviours to be triggered. Ledru is a pairwise similarity approach that selects tests incrementally, adding at each step the test that maximises its dissimilarity to the already selected set, while FAST approximates this process using some big data techniques, such as Shingling, Minhashing, and Locality-Sensitive Hashing, to reduce computational cost. For each approach, we used two types of diversity information. Textual diversity (Ledru-Text [21] and FAST-Text [24]) measures differences in the text of test cases, capturing variations in inputs and method sequences. Bytecode diversity [10] (Ledru-Bytecode and FAST-Bytecode) measures differences in compiled test instructions, capturing variations in actual program behaviours while being more compact and efficient. Comparing these four variants enables us to assess how both the algorithm and the choice of diversity representation impact the ability to kill stubborn mutants. Due to space limitations, more details on the diversity-based algorithms are provided by Elgendy et al. [10].

### C. Effectiveness of Stubborn-Mutant-Killing Tests

We want to identify the impact of stubborn mutants on test suite effectiveness and real fault detection. An interesting observation from Table I, is that some mutants *subsume* others. Just et al. [19] found that subsuming mutants have higher utility, as they are less likely to be equivalent or trivial. Other works found that stubborn mutants are more likely to subsume other mutants [23]. Killing $m_5$ (e.g., by $T_3$ or $T_6$) also results in the killing of $m_1$, $m_4$, and $m_7$. Similarly, killing $m_3$ (via $T_4$) also kills $m_1$, $m_2$, and $m_6$. This subsumption effect suggests that some stubborn mutants may be more valuable than their apparent rarity implies, because they represent faults whose detection indirectly kills multiple other mutants.

Returning to the example, selecting $T_4$ (which kills $m_3$) and either $T_3$ or $T_6$ (which kill $m_5$) is sufficient to kill all seven mutants. This demonstrates the potential utility of stubborn mutants in test suite optimisation. Focusing on tests that can kill the stubborn mutants could not only reduce the test suite size but also improve overall test effectiveness by collapsing redundant mutant coverage.

In our approach, we identify the set of tests that kill the set of stubborn mutants, denoted as ($T_{Stb}$). We analyse the impact of killing the stubborn mutants by analysing the mutation score achieved by $T_{Stb}$. We can identify whether these tests can reveal real faults in the program by calculating the *Fault Reveal Rate* (FRR), which is the proportion of error-revealing tests that appear in $T_{Stb}$. The set of error-revealing tests is denoted as ($T_E$), which we identify using the Defects4J benchmark [18]. The FRR value is in the range $[0, 1]$ and is calculated as $FRR = |T_{Stb} \cap T_E|/|T_E|$, where $FRR = 1$ indicates that all error-revealing tests are included, and $FRR = 0$ that none are.

## IV. Evaluation

To validate the potential of diversity-based test selection as a lightweight means of killing stubborn mutants and to deepen our understanding of their properties, this study is guided by the following research questions:

- **RQ1:** How do different test selection strategies (coverage-based and textual/bytecode diversity-based) compare in their ability to kill stubborn mutants early under different relative stubbornness thresholds?
- **RQ2:** How effectively do the selection strategies detect hard-to-find real faults, and do their behaviours mirror those observed for stubborn mutants?
- **RQ3:** What mutation adequacy do tests that kill stubborn mutants achieve, and how does this relate to real fault detection?
- **RQ4:** Where do stubborn mutants usually occur, and what are their characteristics?

### A. Subjects

Our experiments were conducted on Java projects drawn from Defects4J [18]. From the full benchmark, we selected seven projects for which both JaCoCo [16] and PIT [4] executed without issues. Projects that exhibited persistent

Table II: The test subjects used in our work

| ID | #Versions | #Mutants | # Test Cases | | | |
|---|---|---|---|---|---|---|
| | | | Randoop | EvoSuite | Developer | Total |
| Cli | 5 | 786 | 7425 | 279 | 242 | 7946 |
| Compress | 7 | 1051 | 10769 | 133 | 38 | 10940 |
| Csv | 10 | 767 | 15050 | 260 | 550 | 15860 |
| Jsoup | 16 | 767 | 20575 | 278 | 632 | 21485 |
| Lang | 15 | 4136 | 25893 | 1103 | 688 | 27684 |
| Math | 42 | 10986 | 27235 | 962 | 1612 | 29809 |
| Time | 2 | 268 | 499 | 172 | 69 | 740 |
| Total | 97 | 18826 | 107446 | 3187 | 3831 | 114464 |

coverage-generation or mutation-analysis failures were excluded after extensive troubleshooting efforts. The used projects are listed in Table II, where a total of 97 versions were considered in our experiments. We used PIT [4], which is a widely used mutation tool in academia and industry [33], to generate mutants for fault detection evaluation. For each version of our test subjects, we target specific classes in the code. These classes are specified by Defects4J as target classes that contain real faults.

To ensure our study included a sufficient number of stubborn mutants, we required large and diverse test suites capable of revealing subtle or subsuming behaviours. For this reason, we augmented developer-written tests with automatically generated tests from Randoop [26] and EvoSuite [13]. Larger and more comprehensive test suites increase the likelihood of killing a broad range of mutants, including those that are harder to detect. As noted by Just et al. [18], more complete test suites provide a better approximation when identifying dominator sets (mutants that subsume others), a concept closely related to stubborn mutants.

### B. Methodology

We developed a tool to automate stubborn mutant identification and test selection strategies. We explain data collection, test selection approaches, and stubborn mutant analysis.

*1) Data Collection:* We used publicly available data [11], which includes error-revealing tests, coverage, similarity, and mutation information for the seven projects reported in Table II. The coverage information contains statement coverage, with coverage per test case provided. More details about the format and the structure of the data can be found in the replication package [1].

*2) Test Case Selection Approaches:* As we explained in Section III-A, we used the RSTM model to identify the stubborn mutants. In our experiments, we investigated different thresholds, which are 0.75, 0.5, 0.25, 0.1, 0.05, 0.01, 0.005, 0.003, 0.002, and 0.001. Our tool implemented the six coverage and diversity-based test selection approaches explained in Section III-B. For each stubborn mutant, the tool records the number of the first test case to kill that mutant.

*3) Effectiveness of Stubborn-Mutant-Killing Tests:* For all stubbornness thresholds, the tool determined $T_{Stb}$ (i.e., the set of tests that kill the set of stubborn mutants). Then, it recorded the total mutation score, the mutation score achieved only by $T_{Stb}$, the FRR values, the size of the full test suite, and the size of $T_{Stb}$. To establish a baseline, we also generated random test subsets ($\mathcal{R}$) of the same size as $T_{Stb}$. Each random selection was repeated 31 times, and the results were averaged to reduce variance. This comparison with $\mathcal{R}$ allows us to distinguish the specific effectiveness of stubborn-mutant-killing tests from effects that could arise merely due to test suite size.

Our tool averaged values across each project's versions and counted the number of times a *perfect FRR* (i.e., a value of 1 indicating all error-revealing tests are selected) is achieved.

*4) Stubborn Mutant Characteristics:* We analysed some properties of the most stubborn mutants in each project. The most stubborn mutants have scores ranging from 1 (i.e., killed by only one test) to 5 (i.e., killed by five tests).

For each stubborn mutant, the tool identified the source file and line number where the mutant occurred. The tool parsed the abstract syntax tree to identify the type of the line (e.g., an `if` statement, a `return` statement, a method invocation, etc). The tool also recorded the nesting level of that line. In addition, the tool identified the method containing the mutated line and recorded the access modifier (i.e., `private`, `protected`, `public`), whether it is a `void` or non-`void` method, and whether it is a `static` or instance method. Finally, the tool recorded the mutation operator that generated the mutant. All this information was used to give us an idea of what might cause a mutant to be stubborn.

### C. Threats to Validity

**Construct validity:** Fault difficulty is defined in our study as the number of test cases required to kill a mutant. We acknowledge that a mutant may appear difficult not because it is inherently subtle, but because the affected code region has received less testing attention. To reduce this risk, we augmented developer-written tests with automatically generated tests that systematically explore behaviour and are not influenced by developer' decisions, lowering the likelihood that stubborn mutants are artefacts of limited manual testing.

**Internal validity:** Results could be influenced by how stubborn mutants are identified via RSTM. We also applied the fixed-threshold method from Papadakis et al. [27], observing negligible differences, indicating our findings are robust. Detailed comparisons are available in our replication package [1].

**External validity:** Findings may not generalise beyond our subjects. We mitigated this by evaluating seven diverse Java projects from Defects4J, covering 97 versions and 67 classes, reporting both median results and variability (e.g., standard deviation) with statistical significance (p-values, effect sizes). Project-level results are included in the replication package.

**Reliability:** Reproducibility is critical. We provide a comprehensive replication package [1] with source code, scripts, datasets, and detailed instructions. Methodology and evaluation metrics are fully documented in this paper.

## V. Results

### A. RQ1: Test case selection strategies to kill stubborn mutants

Table III shows our analysis of the test selection strategies in killing the stubborn mutants for all projects in our study. The

Table III: Median and standard deviation of the first test to kill stubborn mutants per selection strategy for all projects

| Threshold Level | #Stb Mutants | Ledru-Text | | Ledru-Bytecode | | FAST-Text | | FAST-Bytecode | | Max-Cov | | Min-Cov | | Max-Cov/FAST-Byte | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Med | SD | Med | SD | Med | SD | Med | SD | Med | SD | Med | SD | **p-value** | **$A_{12}$** |
| 0.75 | 13196 | 16.0 | 332.3 | 14.0 | 253.1 | 50.0 | 599.8 | 20.0 | 330.5 | **5.0** | 308.0 | 444.0 | 686.7 | <**0.0001** | 0.64 (Medium) |
| 0.50 | 12690 | 17.0 | 337.5 | 16.0 | 257.2 | 55.0 | 605.7 | 21.0 | 336.3 | **6.0** | 313.2 | 446.0 | 691.5 | <**0.0001** | 0.64 (Medium) |
| 0.25 | 11111 | 25.0 | 355.8 | 21.0 | 271.9 | 67.0 | 629.7 | 24.0 | 354.6 | **9.0** | 331.5 | 511.0 | 706.0 | <**0.0001** | 0.63 (Small) |
| 0.10 | 8633 | 38.0 | 393.3 | 30.0 | 300.9 | 111.0 | 676.6 | 33.0 | 391.7 | **13.0** | 368.5 | 626.0 | 729.4 | <**0.0001** | 0.61 (Small) |
| 0.05 | 6772 | 55.0 | 427.2 | 40.0 | 328.8 | 167.0 | 714.0 | 39.0 | 423.9 | **17.5** | 395.4 | 716.0 | 737.4 | <**0.0001** | 0.60 (Small) |
| 0.01 | 2649 | 175.0 | 539.7 | 105.0 | 446.2 | 539.0 | 829.2 | 65.0 | 558.8 | **57.0** | 497.7 | 857.0 | 704.2 | <**0.0001** | 0.55 (Small) |
| 0.005 | 1564 | 244.0 | 621.8 | 157.0 | 482.2 | 714.0 | 836.1 | **44.0** | 661.0 | 63.5 | 560.7 | 934.5 | 717.0 | **0.0004** | 0.54 (Small) |
| 0.003 | 1079 | 256.0 | 640.0 | 179.0 | 499.9 | 730.0 | 852.2 | **42.0** | 720.9 | 87.0 | 615.1 | 905.0 | 695.6 | 0.6695 | 0.51 (None) |
| 0.002 | 739 | 361.0 | 695.7 | 235.0 | 549.8 | 758.0 | 914.3 | **61.5** | 828.1 | 128.0 | 688.7 | 1006.0 | 741.7 | 0.5282 | 0.51 (None) |
| 0.001 | 236 | 408.0 | 853.5 | 288.5 | 724.9 | 1664.0 | 1107.1 | **217.5** | 1095.8 | 404.0 | 862.6 | 1601.0 | 814.4 | 0.7467 | 0.49 (None) |

first column shows the stubbornness threshold levels, and the second column shows the number of stubborn mutants that fall in that threshold. The rest of the table shows the median (Med) and standard deviation (SD) of the first test to kill stubborn mutants for every test selection strategy used in our study. The table presents how quickly different test selection strategies are able to kill stubborn mutants, measured by the median number of selected tests needed. Lower medians are better, indicating faster detection. The bolded values highlight the strategy with the lowest median at each threshold level, i.e., the most efficient in killing stubborn mutants early. The relatively large standard deviation values observed across thresholds indicate substantial variability in first-kill positions across subjects and mutants. This suggests that while median trends are informative, performance can vary considerably depending on project characteristics and mutant distribution.

Maximum coverage (Max-Cov) is the best-performing strategy at low stubbornness levels, and starting from threshold level 0.005, it falls behind FAST-Bytecode. Minimum coverage (Min-Cov) is consistently the worst selection strategy, indicating that tests with a narrow focus have no utility in killing stubborn mutants. FAST-Text performed the worst out of all diversity-based strategies. Ledru-Text and Ledru-Bytecode are very close to each other, and their performance is relatively close to that of maximum coverage at broad stubbornness thresholds, and slightly better than FAST-Bytecode. However, after threshold 0.05, they fall behind FAST-Bytecode, and the gap gets bigger with higher threshold levels. FAST-Bytecode was reported by far the best performing in terms of runtime [10]. That makes FAST-Bytecode the best candidate of the diversity-based strategies overall.

To assess whether these observed differences are statistically significant, we conducted Mann-Whitney U tests and the Vargha-Delany effect size on all strategies. Given that six selection strategies yield 15 pairwise comparisons, it is not feasible to discuss all results in detail. Instead, we focus on the two best-performing strategies, FAST-Bytecode and Max-Cov, which also represent the two families of approaches considered in this study: static lightweight diversity-based and dynamic coverage-based.

*Comparison between FAST-Bytecode and Max-Cov:* Although Max-Cov achieves the lowest median number of tests needed to kill stubborn mutants at higher threshold levels (e.g., $\theta$ = 0.75, median = 5), the median values achieved by FAST-Bytecode are close in magnitude. For example, at threshold levels of 0.75, 0.50, and 0.25, FAST-Bytecode yields medians of 20, 21, and 24, respectively. This is significant because FAST-Bytecode is a lightweight strategy, requiring no prior execution of the test suite, whereas Max-Cov depends on dynamic coverage information. The closeness of their performance suggests that a lightweight diversity-based approach (FAST-Bytecode) can be competitive with heavier, execution-dependent approaches in the early stages of test selection.

The last two columns in Table III show the statistical comparison between FAST-Bytecode and Max-Cov using Mann-Whitney U test and the Vargha-Delaney $A_{12}$ effect size. The results show that the performance of Max-Cov is statistically significantly better than FAST-Bytecode for thresholds 0.75 through 0.01, with p-values $\leq$ 0.001, and the effect size shows a medium to small effect. For stricter stubbornness (e.g., $\theta$ $\leq$ 0.005), the statistical difference diminishes, with p-values increasing and $A_{12}$ effect sizes nearing 0.5, implying no practical performance advantage of Max-Cov over FAST-Byte. In fact, from threshold 0.005–0.001, FAST-Bytecode outperforms Max-Cov in terms of median, though the difference is not statistically significant.

**Conclusion for RQ1:** Our findings indicate that while Max-Cov demonstrates strong performance at broader stubbornness thresholds by quickly killing mutants with fewer tests, its advantage decreases as the threshold becomes stricter. FAST-Bytecode is lightweight and does not require pre-execution of the test suite and consistently approaches Max-Cov in early-stage effectiveness and eventually surpasses it in the stricter thresholds (e.g., $\leq$ 0.005). FAST-Bytecode has up to 46% fewer tests required at the strictest threshold than Max-Cov.

### *B. RQ2: Test case selection strategies to detect hard real faults*

Table IV shows our analysis of the test selection strategies in detecting the hard-to-find real faults for all projects in our study. Across all thresholds, the behavioural trends closely

Table IV: Median and standard deviation of the first test to kill hard-faults per selection strategy for all projects

| Threshold Level | #Hard Faults | Ledru-Text | | Ledru-Bytecode | | FAST-Text | | FAST-Bytecode | | Max-Cov | | Min-Cov | | Max-Cov/FAST-Byte | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Med | SD | Med | SD | Med | SD | Med | SD | Med | SD | Med | SD | **p-value** | $\mathbf{A}_{12}$ |
| 0.75 | 94 | 119.5 | 523.1 | 62.5 | 445.9 | 96.0 | 897.8 | 31.0 | 523.5 | **14.5** | 521.9 | 492.5 | 799.5 | 0.0781 | 0.57 (Small) |
| 0.50 | 93 | 129.0 | 525.1 | 63.0 | 447.4 | 110.5 | 900.5 | 32.0 | 525.9 | **14.0** | 524.1 | 518.0 | 803.0 | 0.0718 | 0.58 (Small) |
| 0.25 | 91 | 139.0 | 528.3 | 66.0 | 450.5 | 136.0 | 905.8 | 33.5 | 530.7 | **15.0** | 528.4 | 536.0 | 804.0 | 0.0795 | 0.58 (Small) |
| 0.10 | 86 | 150.0 | 538.2 | 67.5 | 454.7 | 151.0 | 920.5 | 36.0 | 543.8 | **16.5** | 540.0 | 593.0 | 793.1 | 0.1159 | 0.57 (Small) |
| 0.05 | 78 | 178.5 | 555.0 | 87.0 | 466.9 | 190.0 | 933.9 | 43.0 | 566.9 | **21.0** | 560.3 | 605.5 | 794.3 | 0.2220 | 0.56 (Small) |
| 0.01 | 53 | 318.0 | 609.6 | 149.0 | 528.0 | 388.5 | 1028.9 | **58.0** | 664.3 | 96.0 | 639.8 | 807.0 | 814.5 | 0.9795 | 0.50 (None) |
| 0.005 | 38 | 445.0 | 661.8 | 304.0 | 582.4 | 768.0 | 1087.2 | **119.0** | 742.1 | 171.5 | 709.2 | 884.0 | 832.2 | 0.7504 | 0.48 (None) |
| 0.003 | 32 | 504.0 | 706.9 | 304.0 | 614.1 | 920.0 | 1114.3 | **189.0** | 788.7 | 302.5 | 743.1 | 947.5 | 865.4 | 0.7053 | 0.47 (None) |
| 0.002 | 23 | 513.0 | 722.9 | **272.0** | 672.7 | 1003.0 | 1158.2 | 284.0 | 858.7 | 454.0 | 729.4 | 1011.0 | 890.0 | 0.6133 | 0.46 (Small) |
| 0.001 | 8 | 859.0 | 905.6 | 1046.5 | 843.4 | 2508.5 | 1504.5 | **804.5** | 1112.8 | 908.0 | 849.8 | 1783.5 | 1048.8 | 0.8785 | 0.47 (None) |

mirror those observed for stubborn mutants. At broader thresholds (0.75–0.10), Max-Cov consistently identifies and kills faults earliest, with median first-kill positions between 14–21 tests. FAST-Bytecode remains competitive in this region and is consistently the second-best performer, often within 15–20 additional tests.

As the threshold becomes stricter, focusing on the hardest faults, the relative performance shifts. From $\theta \leq 0.01$, FAST-Bytecode surpasses Max-Cov, detecting the hardest real faults earlier (e.g., 58 vs. 96 tests at $\theta = 0.01$). This matches the pattern previously observed for stubborn mutants, where diversity-based selection becomes increasingly advantageous as fault difficulty increases.

However, unlike the stubborn-mutant results, these differences for real faults are not statistically significant across any threshold (all p-values $\geq 0.07$), and the effect sizes remain small or negligible. Thus, while the direction of the trend is consistent, diversity-based selection gains strength as faults become harder to detect, the empirical support is weaker due to the limited number of real faults per threshold and the high standard deviation values.

**Conclusion for RQ2:** The effectiveness of selection strategies on hard-to-find real faults mirrors the patterns observed with stubborn mutants: Max-Cov performs best at broader thresholds, while FAST-Bytecode catches up and surpasses Max-Cov at stricter thresholds. However, due to smaller sample and higher variability in detection cost, these differences are not statistically significant, and effect sizes are small to negligible.

### *C. RQ3: Effectiveness of Stubborn-Mutant-Killing Tests*

Table V summarises the effectiveness of selecting tests that target only stubborn mutants ($T_{Stb}$) at different stubbornness thresholds across all projects. Although our setup considered thresholds from 0.75 down to 0.001, we report only results for thresholds $\geq$0.05. At lower thresholds (0.01–0.001), the number of stubborn mutants was too small to produce meaningful or generalisable results.Thus, for clarity and space, we focus on thresholds yielding consistent, non-trivial results across projects. Full results, including lower thresholds, are available in our replication package [1]. The first column lists the project name, average test suite size, and mutation score (MS) per version. The second column shows the threshold level (TL) used to identify stubborn mutants via RSTM. Columns three and four report the average MS achieved by $T_{Stb}$ and $\mathcal{R}$, a random baseline of equal size. The next two columns show the proportion of versions in which $T_{Stb}$ and $\mathcal{R}$ achieve the same MS as the full suite ("Perfect" MS). Subsequent columns present the average fault-revealing rate (FRR) and the proportion of versions with perfect FRR (i.e., FRR = 1.0). The final column lists the average size of $T_{Stb}$ (equal to $\mathcal{R}$).

Across all projects, targeting only stubborn mutants ($T_{Stb}$) dramatically reduces test suite size. For example, average test counts drop from over 1600 in `Cli` and `Compress` to fewer than 150, and below 50 in `Csv` and `Jsoup`. At a threshold level of 0.75, $T_{Stb}$ consistently achieves high mutation scores, typically matching the full suite with "Perfect MS" near 1.0 across most projects. In contrast, the random baseline $\mathcal{R}$ shows substantially lower MS and fault-revealing performance, with "Perfect MS" and "Perfect FRR" values remaining zero across all thresholds. This confirms that the strong performance of $T_{Stb}$ arises from targeting stubborn mutants rather than from random test reduction.

In terms of real fault detection, FRR varies across projects. Four of the seven subjects (`Csv`, `Jsoup`, `Lang`, and `Math`) show moderate to high FRR at threshold 0.75, indicating that many real fault-revealing tests are captured within $T_{Stb}$. In contrast, `Cli` and `Compress` exhibit consistently low FRR, and the `Time` project highlights a limitation: none of its real fault-revealing tests were included in $T_{Stb}$ across the two evaluated versions (FRR = 0). However, these three projects contain relatively few real faults (five, seven, and two for `Cli`, `Compress`, and `Time`, respectively), limiting the representativeness of their FRR outcomes.

A decline in FRR is therefore most evident in projects with sparse real fault data. Conversely, systems with larger fault sets exhibit consistently higher FRR, suggesting that $T_{Stb}$ is more effective when sufficient real faults are available for evaluation.

Overall, the results show that stubborn mutants serve as valuable indicators of test effectiveness. Focusing on tests that detect them—particularly early in testing—can improve both adequacy and cost-efficiency. Empirically, a threshold of 0.75 offers a strong balance between FRR and suite size across all studied projects. We thus recommend 0.75 as a practical

Table V: Effectiveness analysis of stubborn mutants for all projects

| Project (Avg Size / Avg MS) | TL | Avg MS | | Perfect MS | | Avg FRR | | Perfect FRR | | Avg Size |
|---|---|---|---|---|---|---|---|---|---|---|
| | | $T_{Stb}$ | $\mathcal{R}$ | $T_{Stb}$ | $\mathcal{R}$ | $T_{Stb}$ | $\mathcal{R}$ | $T_{Stb}$ | $\mathcal{R}$ | |
| Cli (1738 / 0.9) | 0.75 | 0.9 | 0.85 | 1.0 | 0.0 | 0.68 | 0.40 | 0.5 | 0.0 | 141 |
| | 0.50 | 0.9 | 0.84 | 1.0 | 0.0 | 0.68 | 0.38 | 0.5 | 0.0 | 136 |
| | 0.25 | 0.9 | 0.82 | 1.0 | 0.0 | 0.64 | 0.37 | 0.3 | 0.0 | 125 |
| | 0.10 | 0.9 | 0.76 | 1.0 | 0.0 | 0.63 | 0.24 | 0.0 | 0.0 | 107 |
| | 0.05 | 0.9 | 0.73 | 1.0 | 0.0 | 0.5 | 0.16 | 0.0 | 0.0 | 95 |
| Compress (1617 / 0.72) | 0.75 | 0.72 | 0.54 | 1.0 | 0.0 | 0.63 | 0.33 | 0.2 | 0.0 | 112 |
| | 0.50 | 0.72 | 0.56 | 1.0 | 0.0 | 0.63 | 0.33 | 0.2 | 0.0 | 108 |
| | 0.25 | 0.70 | 0.46 | 0.6 | 0.0 | 0.63 | 0.24 | 0.2 | 0.0 | 104 |
| | 0.10 | 0.69 | 0.34 | 0.4 | 0.0 | 0.43 | 0.06 | 0.2 | 0.0 | 98 |
| | 0.05 | 0.69 | 0.31 | 0.4 | 0.0 | 0.43 | 0.03 | 0.2 | 0.0 | 97 |
| Csv (1902 / 0.69) | 0.75 | 0.69 | 0.64 | 0.8 | 0.0 | 0.73 | 0.38 | 0.6 | 0.0 | 32 |
| | 0.50 | 0.69 | 0.63 | 0.8 | 0.0 | 0.73 | 0.38 | 0.6 | 0.0 | 32 |
| | 0.25 | 0.69 | 0.62 | 0.6 | 0.0 | 0.53 | 0.27 | 0.4 | 0.0 | 27 |
| | 0.10 | 0.68 | 0.50 | 0.6 | 0.0 | 0.53 | 0.06 | 0.4 | 0.0 | 20 |
| | 0.05 | 0.63 | 0.44 | 0.4 | 0.0 | 0.53 | 0.02 | 0.4 | 0.0 | 19 |
| Jsoup (1798 / 0.86) | 0.75 | 0.83 | 0.67 | 0.9 | 0.1 | 0.82 | 0.31 | 0.8 | 0.0 | 51 |
| | 0.50 | 0.83 | 0.67 | 0.9 | 0.0 | 0.82 | 0.27 | 0.8 | 0.0 | 49 |
| | 0.25 | 0.83 | 0.56 | 0.9 | 0.0 | 0.72 | 0.15 | 0.7 | 0.0 | 47 |
| | 0.10 | 0.83 | 0.54 | 0.9 | 0.0 | 0.72 | 0.11 | 0.7 | 0.0 | 43 |
| | 0.05 | 0.79 | 0.49 | 0.8 | 0.0 | 0.6 | 0.06 | 0.5 | 0.0 | 40 |
| Lang (1811 / 0.84) | 0.75 | 0.84 | 0.77 | 1.00 | 0.27 | 0.88 | 0.46 | 0.82 | 0.0 | 268 |
| | 0.50 | 0.77 | 0.68 | 0.82 | 0.18 | 0.78 | 0.39 | 0.73 | 0.0 | 259 |
| | 0.25 | 0.76 | 0.67 | 0.73 | 0.18 | 0.72 | 0.37 | 0.64 | 0.0 | 246 |
| | 0.10 | 0.67 | 0.57 | 0.64 | 0.09 | 0.67 | 0.25 | 0.64 | 0.0 | 202 |
| | 0.05 | 0.58 | 0.50 | 0.45 | 0.00 | 0.58 | 0.16 | 0.55 | 0.0 | 155 |
| Math (625 / 0.82) | 0.75 | 0.82 | 0.73 | 0.94 | 0.10 | 0.84 | 0.43 | 0.74 | 0.0 | 157 |
| | 0.50 | 0.79 | 0.66 | 0.90 | 0.06 | 0.81 | 0.38 | 0.71 | 0.0 | 151 |
| | 0.25 | 0.78 | 0.64 | 0.84 | 0.06 | 0.77 | 0.32 | 0.68 | 0.0 | 125 |
| | 0.10 | 0.73 | 0.57 | 0.61 | 0.00 | 0.69 | 0.20 | 0.61 | 0.0 | 90 |
| | 0.05 | 0.64 | 0.44 | 0.35 | 0.00 | 0.54 | 0.11 | 0.48 | 0.0 | 69 |
| Time (370 / 0.8) | 0.75 | 0.80 | 0.72 | 1.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 102 |
| | 0.50 | 0.80 | 0.70 | 1.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 98 |
| | 0.25 | 0.80 | 0.65 | 1.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 83 |
| | 0.10 | 0.80 | 0.62 | 1.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 64 |
| | 0.05 | 0.78 | 0.50 | 0.5 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 34 |

default, while allowing practitioners to adjust it according to project scale and testing goals.

**Conclusion for RQ3:** Stubborn mutants effectively guide targeted test selection. At moderate relative thresholds (e.g., 0.75), focusing only on stubborn mutants achieves high mutation adequacy and strong FRR while substantially reducing test suite size. Across our subjects, tests that killed only stubborn mutants comprised just 2–25% of the full suite, yet achieved perfect mutation scores in all but one project. In six of seven cases, these subsets reached average real fault FRR values between 63% and 88%, showing that many fault-revealing tests are included when focusing on stubborn mutants. By contrast, randomly selected subsets of equal size consistently underperformed—yielding lower mutation scores, lower FRR, and never achieving perfect MS or FRR.

### *D. RQ4: Characteristics of stubbornness*

To examine whether certain code characteristics are associated with stubborn mutants, we measured the proportion of stubborn mutants with each characteristic compared to all mutants and performed Chi-Square tests of independence [28]. Categories included access modifiers, method types, node types, mutation operators, and nesting levels, with count data indicating how many mutants belonged to each stubbornness score (Score1–Score5). Chi-Square tests assess whether a category is significantly associated with being a top-score mutant. We report both statistical significance ($p$-value) and effect size via Cramér's V [5] ( 0.1 = small, 0.3 = medium, 0.5 = large effect).

Since high proportions do not necessarily imply statistical significance, we also use standardised residuals (z-scores) to measure deviations from expected counts, which also account for sample size. A higher z-score indicates a stronger over- or underrepresentation, accounting for sample size, with $|z| > 2$ considered significant.

Table VI shows proportions and statistical results for each characteristic, and Figure 3 visualises the distributions.

Prior work [33], [31] examined mutation score, code testability, and operator class correlations with stubborn mutants. The mutant operator class ABS is not included in PIT's default operators, and some PIT operators are method-level, beyond the five general classes. It would be interesting to explore links between code observability and PIT operators in future work.

Across all stubbornness scores, Chi-Square tests confirmed significant associations ($p < 0.001$, Table VI), though effect sizes were small (Cramér's V: 0.08–0.13), indicating subtle but meaningful trends.

*1) Access Modifier:* Stubborn mutants, particularly at higher scores, are more likely in *private* methods and less likely in *public* or *protected* ones (Figure 3a). Private methods show strong overrepresentation in top scores (e.g., Score1 $z = 12.0$), while protected methods are consistently underrepresented. This indicates that reduced visibility may limit observability, making mutants harder to detect, consistent with Zhu et al. [33]'s recommendation to refactor private methods to increase test effectiveness.

*2) Method Type: Void* methods are more prone to stubborn mutants across all scores, particularly Score1 ($z = 10.6$), while *non-void* methods are underrepresented, likely because return values aid assertion-based testing. Constructors show moderate overrepresentation, but the trend is strongest for void methods (Figure 3a). This aligns with Zhu et al. [33]'s suggestion to refactor void methods to improve testability.

*3) Mutation Operator:* Some operators disproportionately produce stubborn mutants. *VoidMethodCallMutator*, *MathMutator*, and *EmptyObjectReturnValsMutator* are consistently overrepresented, suggesting subtle semantic changes that evade detection, while *NegateConditionalsMutator* and *NullReturnValsMutator* are underrepresented (Figure 3b). Although Yao et al. [31] highlighted LCR as prone to stubborn mutants, we found *BooleanTrueReturnValsMutator* and

Table VI: Proportions and statistical analysis of each category across the top five scored subsets. Bold values indicate the highest proportions and standardised residuals (z-scores) within each category for the corresponding subset. $p$ denotes the $p$-value, $C$ is Cramér's V (effect size), $z$ represents standardised residuals, and % indicates the proportion of the subset. An asterisk (*) denotes $p < 0.001$.

| Category | Value | Score1 | | | | Score2 | | | | Score3 | | | | Score4 | | | | Score5 | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | % | $z$ | $p$ | $C$ | % | $z$ | $p$ | $C$ | % | $z$ | $p$ | $C$ | % | $z$ | $p$ | $C$ | % | $z$ | $p$ | $C$ |
| Access Modifier | public | 8.1 | -7.0 | | | 16.2 | -3.6 | | | 21.9 | -1.5 | | | 26.7 | 0.3 | | | 30.2 | 1.7 | | |
| | **private** | **17.2** | **12.0** | * | 0.13 | **23.8** | **8.5** | * | 0.11 | **27.2** | **5.8** | * | 0.09 | **29.5** | **3.6** | * | 0.08 | **30.8** | **2.1** | * | 0.08 |
| | protected | 7.4 | -4.0 | | | 11.0 | -6.3 | | | 14.3 | -6.7 | | | 16.9 | -7.1 | | | 18.3 | -7.6 | | |
| | package-private | 11.0 | 0.1 | | | 19.4 | 0.7 | | | 24.6 | 0.8 | | | 28.5 | 0.8 | | | 30.6 | 0.6 | | |
| Method Type | Non-void | 8.7 | -6.3 | | | 15.6 | -5.7 | | | 20.6 | -4.4 | | | 25.1 | -2.8 | | | 27.9 | -2.1 | | |
| | **Void** | **17.4** | **10.6** | * | 0.12 | **25.4** | **9.3** | * | 0.11 | **29.0** | **6.9** | * | 0.09 | **30.8** | **4.5** | * | 0.06 | **32.4** | **3.3** | * | 0.04 |
| | Constructor | 14.1 | 2.1 | | | 23.1 | 2.6 | | | 28.5 | 2.6 | | | 30.2 | 1.6 | | | 31.9 | 1.1 | | |
| Mutator | Negate-Conditionals | 8.6 | -4.5 | | | 14.6 | -5.3 | | | 19.7 | -4.2 | | | 24.3 | -2.9 | | | 28.1 | -1.1 | | |
| | Math | 14.0 | 5.8 | | | 22.8 | **6.6** | | | 27.3 | **5.7** | | | 30.7 | 4.7 | | | 32.4 | 3.7 | | |
| | Conditionals-Boundary | 7.8 | -3.6 | | | 12.3 | -5.3 | | | 14.6 | -6.6 | | | 16.9 | -7.3 | | | 18.4 | -7.7 | | |
| | NullReturnVals | 5.8 | -4.8 | | | 12.6 | -4.0 | | | 16.8 | -3.9 | | | 19.1 | -4.5 | | | 20.4 | -5.0 | | |
| | **VoidMethodCall** | 19.2 | **7.4** | | | 26.9 | **6.1** | | | 32.1 | **5.7** | | | **36.0** | **5.4** | | | 37.9 | -3.1 | | |
| | Primitive-ReturnsCall | 10.0 | -0.6 | * | 0.12 | 16.3 | -0.9 | * | 0.13 | 23.6 | 0.4 | * | 0.13 | 27.8 | 0.6 | * | 0.13 | 29.9 | 0.4 | * | 0.13 |
| | EmptyObject-ReturnVals | 8.3 | -1.5 | | | 26.4 | 3.9 | | | **33.2** | 4.3 | | | **36.5** | 3.8 | | | **41.5** | **4.5** | | |
| | BooleanTrue-ReturnVals | 9.9 | -0.5 | | | 16.5 | -0.7 | | | 22.4 | -0.1 | | | 29.0 | 0.9 | | | 30.1 | 0.4 | | |
| | Increments | **21.3** | 4.5 | | | **27.4** | 3.1 | | | 30.0 | 2.1 | | | 32.5 | 1.6 | | | 35.5 | 1.7 | | |
| | BooleanFalse-ReturnVals | 9.5 | -0.5 | | | 20.1 | 0.6 | | | 27.8 | 1.4 | | | **36.7** | 2.6 | | | 40.2 | 2.7 | | |
| | InvertNegs | 18.0 | 2.4 | | | 21.3 | 0.9 | | | 27.9 | 1.2 | | | 33.6 | 1.5 | | | 33.6 | 0.9 | | |
| Nesting Level | 0 | 8.3 | -6.2 | | | 13.3 | -8.8 | | | 17.6 | -8.6 | | | 21.0 | -8.4 | | | 23.6 | -7.9 | | |
| | 1 | 10.8 | 0.1 | | | 20.1 | 3.0 | | | 25.8 | 3.8 | | | 29.6 | 3.7 | | | 31.9 | 3.2 | | |
| | 2 | 14.1 | 4.0 | | | 24.3 | **5.9** | | | 29.0 | **5.2** | | | 32.4 | **4.6** | | | 35.2 | **4.6** | | |
| | **3** | 17.6 | **5.8** | * | 0.11 | 23.8 | 3.9 | * | 0.13 | 29.6 | 4.1 | * | 0.13 | 35.3 | **4.9** | * | 0.13 | 37.4 | **4.4** | * | 0.12 |
| | **4** | **19.7** | **5.4** | | | **30.1** | **5.7** | | | **33.1** | **4.4** | | | 36.6 | 4.0 | | | 39.1 | 3.8 | | |
| | 5 | **20.4** | 3.7 | | | 29.3 | 3.4 | | | **33.1** | 2.7 | | | 37.6 | 2.7 | | | 41.4 | 2.9 | | |
| | 6 | 17.9 | 1.8 | | | **29.9** | 2.3 | | | **32.8** | 1.7 | | | **38.8** | 2.0 | | | **44.8** | 2.4 | | |
| Node Type | BinaryOperation | 13.6 | 5.6 | | | 22.1 | 6.2 | | | 26.5 | 5.1 | | | 29.6 | 4.0 | | | 32.2 | 3.8 | | |
| | IfStatement | 8.0 | -5.6 | | | 13.4 | -7.0 | | | 17.9 | -6.6 | | | 22.4 | -5.2 | | | 25.3 | -4.5 | | |
| | ReturnStatement | 7.9 | -4.4 | | | 16.6 | -1.7 | | | 22.5 | -0.3 | | | 26.5 | 0.0 | | | 28.7 | -0.4 | | |
| | ForStatement | 8.1 | -2.5 | | | 13.8 | -2.9 | | | 17.3 | -3.4 | | | 19.4 | -4.1 | | | 21.4 | -4.2 | | |
| | MethodInvo. | 20.7 | **8.8** | * | 0.13 | 28.3 | **7.1** | * | 0.12 | 33.5 | **6.6** | * | 0.12 | 37.3 | **6.2** | * | 0.11 | 39.3 | **5.6** | * | 0.10 |
| | TernaryExp. | 17.5 | 2.2 | | | 21.9 | 1.0 | | | 29.8 | 1.6 | | | 36.0 | 2.0 | | | **42.1** | 2.6 | | |
| | StatementExp. | 3.5 | -1.7 | | | 6.9 | -2.0 | | | 10.3 | -2.0 | | | 17.2 | -1.4 | | | 17.2 | -1.7 | | |
| | WhileStatement | 26.3 | 2.1 | | | **36.8** | 1.9 | | | **42.1** | 1.8 | | | **42.1** | 1.3 | | | **42.1** | 1.1 | | |
| | VariableDeclare | **28.6** | 2.0 | | | 28.6 | 0.9 | | | 35.7 | 1.0 | | | 35.7 | 0.7 | | | 35.7 | 0.5 | | |

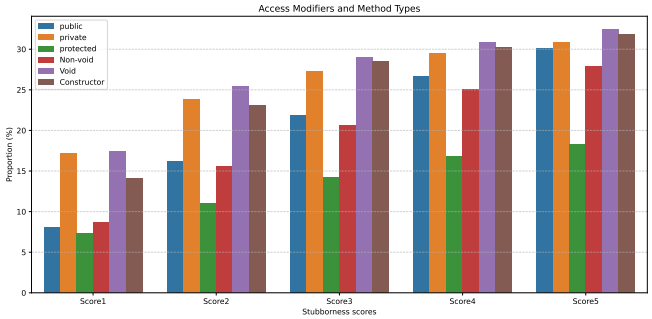
*BooleanFalseReturnValsMutator* (both LCR) near baseline ($z = -0.5$), aligning with expectations.

*4) Nesting Level:* Mutants in deeper code (levels 3–5) are more likely to be stubborn, peaking at level 3 for Score1 ($z = 5.8$), while flat code (level 0) is underrepresented, supporting the idea that structural complexity increases mutation resilience. Higher nesting levels beyond 3 are not overrepresented, likely due to small sample sizes (Figure 3c).

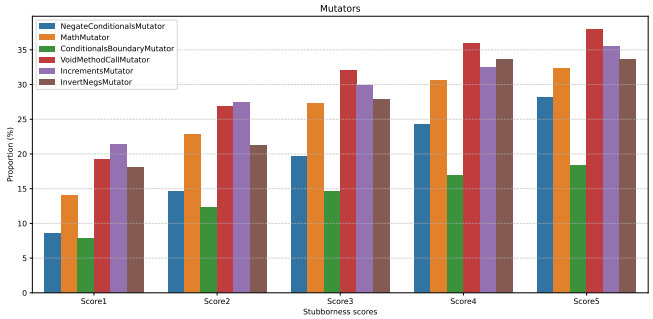
*5) Node Type:* Stubborn mutants cluster in specific constructs. *MethodInvocation* and *BinaryOperation* nodes are consistently overrepresented, especially at higher scores (e.g., *MethodInvocation* Score1: $z = 8.8$), whereas control-flow nodes (*IfStatement*, *ReturnStatement*, *ForStatement*) are underrepresented, suggesting mutations in branching logic are easier to detect (Figure 3d).

While our analysis shows strong associations between certain code characteristics and stubborn mutants, explaining why they occur preferentially in some constructs is beyond this study and left for future work.

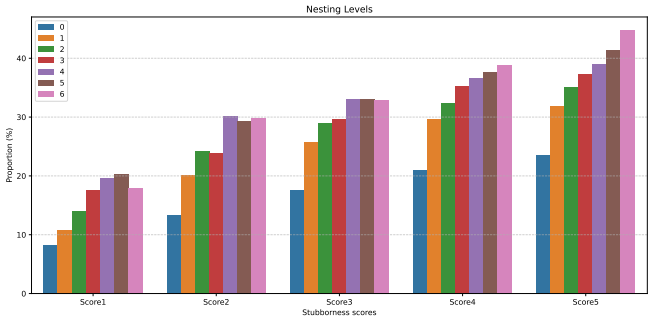
**Conclusion for RQ4:** Our results demonstrate clear links between code characteristics and mutant stubbornness: **Access modifiers:** *Private* methods are more likely to contain stubborn mutants, with over-representation decreasing from 12 (Score1) to 3.3 (Score5), likely due to limited visibility reducing test observability. **Method return types:** *Void* methods are prone to stubborn mutants (10.6 at Score1 down to 3.3 at Score5), as the absence of return values hampers assertion-based testing. **Mutation**



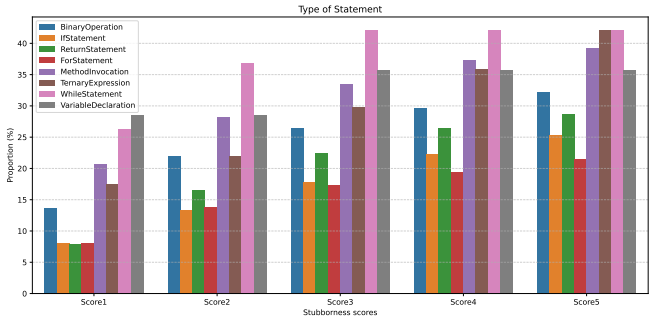

(a) Access Modifiers and Method Types




(b) Mutators




(c) Nesting Levels




(d) Type of Statement

Figure 3: Bar plots showing the distribution of stubborn mutants proportions across different categories and ranks.

**operators:** Method-level operators generate more stubborn mutants, over-represented 7.4 at Score1 and still 5.4 at Score4, indicating harder detection. **Nesting depth:** Mutants in deeper code (levels 3–4) are over-represented (5.8→3.9 for level 3, 5.7→3.8 for level 4), reflecting increased structural complexity and reduced control-path observability.

## VI. Related Work

The link between equivalent and stubborn mutants was studied by Yao et al. [31], finding certain mutation operator classes, such as LCR, are more likely to produce stubborn mutants. While their focus was on operator classes, we also consider code-level characteristics such as access modifiers, return types, and nesting depth.

Zhu et al. [33] investigated the relationship between mutation score and code testability and observability, showing that metrics like test directness strongly influence killability and suggesting refactorings to improve scores. Our work complements this by empirically identifying code attributes that correlate with stubbornness across diverse subjects. Papadakis et al. [27] advocated prioritising mutants that correlate with real fault revelation, showing that hard-to-kill mutants may better indicate real faults than subsuming mutants. We build on this insight by using the RSTM to systematically identify stubborn mutants and evaluate their practical utility in test selection and fault reveal rate.

Shin et al. [30] proposed a diversity-aware mutation adequacy criterion to improve fault detection, but it does not focus on stubborn mutants. Our work specifically characterises stubborn mutants, quantifies their fault-revealing potential, and evaluates how test selection strategies target them effectively. Chekam et al. [3] used dynamic symbolic execution, Lin et al. [22] applied ranking strategies, and Dang et al. [6] employed co-evolutionary genetic algorithms to detect stubborn mutants more efficiently. These approaches improve test generation by targeting hard-to-kill mutants, whereas we focus on mutants already killable but killed by very few tests, evaluating their value for selecting efficient subsets from existing test suites using lightweight, scalable techniques.

Recent ML-based approaches also target mutant selection. Chekam et al. [2] used feature engineering to target stubborn mutants, and Cerebro [14] employs an encoder-decoder to predict subsuming mutants. While effective, these methods rely on prior mutant labels, whereas we evaluate diversity- and coverage-based test selection strategies that are agnostic to stubborn mutants, using RSTM only as a benchmark.

## VII. Conclusions and Future Work

This paper presented the first empirical study examining the relationship between test selection strategies and fault difficulty, using stubborn mutants as proxies for difficult faults and also hard-to-find real faults. We compared dynamic coverage-based and lightweight, static diversity-based techniques across seven real-world Java projects. Results show that coverage-based approaches perform well at broader thresholds, but under stricter definitions of stubbornness, bytecode-based

FAST outperforms them, requiring up to 46% fewer tests. We also observed strong correlations between mutant stubbornness and code properties, including access modifiers, return types, mutation operator categories, and nesting depth. Tests that exclusively kill stubborn mutants constitute only 2–25% of a suite, yet achieve full mutation adequacy and up to 88% fault-revealing rate.

For future work, we plan to explore stubborn mutant detection using semantic models such as Reachability-Infection-Propagation (RIP) to capture deeper execution characteristics. We will further investigate the relationship between stubborn mutants and testability, examining structural factors (e.g., control-flow constructs) that influence fault-detection difficulty. Methodologically, we aim to refine threshold selection by reducing reliance on $S_{max}$, which may be sensitive to outliers, and instead consider aggregate statistics such as average killability. We also plan to evaluate weighted APFD to account for fault criticality (e.g., severity or impact) and incorporate cost-aware metrics reflecting test execution time. Finally, we intend to extend experiments to additional systems and languages, and provide qualitative case studies illustrating which tests kill stubborn mutants and why diversity-based approaches are more effective. Combining RSTM with ML-based pipelines may further enable context-sensitive stubbornness measures.

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